20m CW Transmitter — Design Summary

Push-pull 6146B PA, 12HG7 driver, 14.2 MHz, ~51W output into 50 Ω

Simulation target: QUCS-S / ngspice. Date: 2026-05-29.

Signal Chain

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[V1,V2 exciter] → [12HG7 push-pull driver] → [driver output transformer]
  → [6146B push-pull PA] → [PA output transformer (L4/L5/L6)]
  → [6:1 balun (300 Ω balanced → 50 Ω unbal)]
  → [5-pole Chebyshev LPF, fc=18 MHz]
  → [50 Ω output to external antenna tuner]
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Operating Point

Parameter	Value	Notes
Carrier frequency	14.2 MHz	20m band
Plate supply Vp	600 V	PA V5, fixed source
Screen supply Vs	200 V	PA SpicePar SCREEN
Grid bias Vg	-85 V	PA SpicePar GRID, near-Class C
Driver supply	275 V	V3 in xmitter.sch
Nominal V_IN drive	3 V peak	For ~51 W output
Output power (nominal)	~51 W	into 50 Ω at 14.2 MHz
Output power (full drive ceiling)	~55 W	V_IN=5V at safe Pp limit
Key-up idle plate current	0 mA	Class C cutoff

Per-Tube Operating Parameters (6146B at nominal drive)

Quantity	Value	Datasheet rating
DC plate current (key down)	~37 mA	150 mA max
Plate dissipation	~18 W	25 W max
DC screen current	~3 mA	15 mA max
Screen dissipation	~0.6 W	3 W max
Plate efficiency	~59%	Class C theoretical 70-80%

Component Values by Stage

Driver Stage (Driver_subcircuit.sch)

Component	Value	Function
SUB1, SUB2 (driver tubes)	12HG7 (Koren model)	Push-pull pentode driver pair
R1, R8 (PA grid stoppers)	100 Ω	Parasitic suppression
R3, R9 (driver cathode)	100 Ω	Cathode self-bias resistor
R2, R6 (driver screen)	1 kΩ	Screen series resistor
R4, R7 (grid leak)	47 kΩ	Grid leak to bias supply
C2, C4 (cathode bypass)	0.01 µF	Cathode RF bypass
C5, C6, C7 (decoupling)	0.01 µF	Power-rail RF decoupling
V1, V5 (screen supply)	150 V	Driver screen B+
V2, V6 (plate supply)	275 V	Driver plate B+
V11 (driver grid bias)	-3 V	Driver grid bias
L1, L3 (plate inductors)	1 µH	Plate RF chokes
L2, L4 (B+ chokes)	1 mH	B+ feed chokes

Driver Output Transformer (Driver_output_transformer_subcircuit.sch)

Component	Value	Function
INPUT_L (primary halves)	4.8 µH each	Primary winding, balanced
OUTPUT_L (secondary halves)	1.14 µH each	Secondary, balanced
C_IN (primary tuning)	6.0 pF	Tunes peak to 14.2 MHz
C3, C4 (output coupling)	1000 pF	DC blocking caps to PA grids
K (coupling factor)	0.95 (all pairs)	Cross-coupled primaries/secondaries

PA Stage (PA_subcircuit.sch)

Component	Value	Function
SUB1, X1 (PA tubes)	6146B (Koren model)	Push-pull beam power tetrode pair
V5 (plate supply)	600 V	PA B+
V2, V4 (screen supply)	200 V (=SCREEN)	Screen B+
V1, V3 (grid bias)	-85 V (=-GRID)	Class C grid bias
R1, R5 (cathode return)	10 Ω	Cathode bias / current sense
R2, R6 (grid stopper)	220 Ω	HF parasitic suppression
R3, R7 (screen series)	1 kΩ	Screen current limit
R4, R8 (grid leak)	22 kΩ	Grid leak to bias supply
R14, R15 (anti-parasitic)	47 Ω	Plate VHF parasitic suppression

C6, C7 (neutralizing)	0.24 pF	Cross-coupled neutralization caps
C8, C10 (screen bypass)	0.01 µF	Screen RF bypass
C9, C11 (screen tuning)	100 pF	Screen RF tuning
C12, C13 (plate tank, C_TANK)	33 pF each	Plate-to-plate tank capacitor halves
L4, L5 (plate inductors)	1.71 µH each	Primary tank halves, K_45=0.75
L6 (output secondary)	0.27 µH	Output transformer secondary, K_46=K_56=0.80
L10, L11 (RF feed)	1 µH each	RF feed isolation

Output Balun (Balun_6to1_subcircuit.sch)

6:1 impedance step-down, 300 Ω balanced primary to 50 Ω unbalanced secondary. Voltage ratio $\sqrt{6} \approx 2.45$. In a real build, implement as Guanella-style transmission-line transformer on a #43 or #61 ferrite core, or two interleaved bifilar windings.

Component	Value	Function
LP (primary)	14 µH	Balanced primary, sees ~300 Ω differential from PA
LS (secondary)	2.33 µH	Single-ended secondary, drives 50 Ω load
K (coupling)	0.95	Realistic tight coupling

Low-Pass Filter (LPF_50ohm_subcircuit.sch)

5-pole 0.5 dB Chebyshev LPF, cutoff 18 MHz, Z0 = 50 Ω single-ended. Provides harmonic rejection downstream of the balun. At 14.2 MHz the LPF passes essentially unattenuated; at 28.4 MHz (2nd harmonic) ~30 dB attenuation; at 42.6 MHz (3rd) ~50 dB.

Component	Value	Position
C1	300 pF	Shunt to GND at input
L1	540 nH	Series between IN and node1
C2	450 pF	Shunt to GND at node1 (middle)
L2	540 nH	Series between node1 and OUT
C3	300 pF	Shunt to GND at OUT

Top-Level (xmitter.sch)

Component	Value	Function
V1, V2 (excitation)	{V_IN} V peak, 14.2 MHz	Push-pull driver grid drive
V3 (driver B+)	275 V	Driver plate supply (via driver tank CT)
R1 (load)	50 Ω	Antenna tuner / external load
R7, R8 (grid leak)	1 MΩ	DC return for driver grid AC sources
SpicePar V_IN	3 V nominal	Drive amplitude (range 0 - 5 V usable)

Verified Performance: V_IN sweep through full chain (PA → balun → LPF → 50 Ω load)

Simulated at Vp=600V, Vs=200V, Vg=-85V, R_load=50 Ω. Measured by ngspice tran with uic initialization, 600 µs settling, RMS over last 10 µs.

V_IN	P_out	Ip/tube	Pp/tube	Is/tube	Ps/tube	Eff	Notes
0 V	0.0 W	0 mA	0 W	0 mA	0.00 W	idle	KEY UP (Class C cutoff)
1 V	0.5 W	7 mA	4.1 W	0.6 mA	0.12 W	6%	below threshold
2 V	39.4 W	66 mA	20.2 W	6.1 mA	1.23 W	48%	startup
3 V	**51.9 W**	**76 mA**	**19.7 W**	**8.2 mA**	**1.63 W**	**55%**	**NOMINAL — all safe**
4 V	53.9 W	78 mA	19.6 W	8.6 mA	1.72 W	56%	safe

5 V	54.9 W	89 mA	25.7 W	9.0 mA	1.79 W	50%	Pp AT LIMIT — max drive
7 V	58.4 W	116 mA	40.2 W	9.5 mA	1.90 W	41%	Pp OVER rating — DO NOT USE
10 V	62.5 W	130 mA	46.6 W	9.9 mA	1.99 W	39%	Pp OVER — DO NOT USE
15 V	61.9 W	144 mA	55.5 W	10.5 mA	2.10 W	35%	Ip and Pp OVER — DO NOT USE

Ratings recap (per tube): Pp_max = 25 W, Ip_max = 150 mA, Ps_max = 3 W, Is_max = 15 mA. Values shown with no warning are within rating; rows in red exceed Pp.

Notes

Nominal operating point: V_IN = 3 V gives ~51 W output with all tube parameters well within ratings (Pp 19.7 W of 25 W max, Ps 1.63 W of 3 W max).

Maximum drive: do not exceed V_IN = 5 V. Beyond this, plate dissipation exceeds the 25 W per-tube rating, risking tube damage in a real build (simulator does not enforce thermal limits).

Variable output: output is approximately quadratic in V_IN above the Class C threshold (~2 V). Use a drive-level control on the exciter for QRP / variable power operation.

Class C harmonic content: push-pull cancels even harmonics (2nd, 4th, 6th ≈ -60 dBc in simulation; -25 to -35 dBc in real-world build due to tube mismatch). 3rd harmonic is the largest residual (~-43 dBc). Combined with the 50 Ω LPF (~50 dB attenuation at 42.6 MHz), the output spectrum easily meets FCC Part 97 limits.

Glossary of Abbreviations and Symbols

Quick reference for the symbols and acronyms used throughout this document. Tube-electrode subscripts: p = plate (anode), s = screen, g = grid.

Voltages

Symbol	Meaning	Value in this design
Vp	Plate (anode) DC supply voltage	600 V
Vs (SCREEN)	Screen grid DC supply voltage	200 V
Vg (GRID)	Control grid DC bias voltage (negative)	-85 V
V_IN	Exciter drive amplitude at xmitter.sch top level (peak, sinusoidal)	0 - 5 V usable
CT	Center-tap of a transformer winding (often B+ feed point)	—

Currents, powers, efficiency

Symbol	Meaning	Computed as
Ip	DC plate current per tube	I(V5_supply) / 2 (V5 feeds both plates via CT)
Is	DC screen current per tube	I(screen supply) for that tube
Ig	DC grid current per tube (Class C: small, positive during conduction)	—
Pin	Total DC input power (plate supply only)	Vp × Ip_total = Vp × 2 × Ip(per tube)
P_out	RF output power delivered to R1 (50 Ω antenna terminal)	V_R1_rms² / 50
Pp / tube	Plate dissipation per tube (heat at the plate)	(Pin − P_out) / 2 ; symmetric tubes
Ps / tube	Screen dissipation per tube	Vs × Is
Eff	Plate efficiency: how much of DC input becomes RF output	P_out / Pin × 100%

Circuit elements and concepts

Term	Meaning
Class C	Amplifier biased deep into cutoff (Vg well below the tube's cutoff voltage). Tubes conduct only on narrow positive grid peaks. Highest ef
Push-pull	Two tubes operating 180° out of phase, one driving each half-cycle. Even-order harmonics (2nd, 4th, ...) cancel at the output transforme
Balun	Balanced-to-unbalanced transformer. Here used as 6:1 impedance step-down (300 Ω balanced PA output → 50 Ω single-ended coaxial
LPF	Low-pass filter. Here a 5-pole 0.5 dB Chebyshev with fc = 18 MHz. Passes 14.2 MHz with <0.5 dB loss; attenuates 28.4 MHz (2nd) by ~
K (coupling factor)	Magnetic coupling between two windings of a transformer. K=1.0 is perfect coupling (impossible in practice). Real ferrite transformers a
L_pp	Plate-to-plate inductance of the PA primary tank (= L4 + L5 + 2·M, where M = K·√(L4·L5)). For our PA: 1.71 + 1.71 + 2·(0.75·1.71) = 5.9
R_pp	Plate-to-plate AC load impedance seen by the tubes. For Class C max power, optimal R_pp ≈ (Vp−Vsat)² / (2·P_out) — here about 4 k
Schade formula	P_max = (Vp − V_saturation)² / (2 · R_load). Estimates max plate-circuit output power for a given supply voltage, tube saturation voltag
Koren model	A SPICE pentode/triode tube model formulation by Norman Koren (1996) that uses a smooth analytic E1 = (Vp/KP) · ln(1 + exp(KP·X))
uic	ngspice .tran option: "use initial conditions". Skips the DC operating-point analysis at t=0 and starts the transient simulation directly from
QRP	Amateur radio shorthand for low-power operation (typically ≤ 5 W output for CW). This rig supports QRP by reducing V_IN below the no

dBc	Decibels relative to the fundamental (carrier). A 2nd-harmonic level of -43 dBc means the 2nd harmonic is 43 dB below the fundamenta
fc	Cutoff frequency of a filter. For Chebyshev LPF, fc is the -3 dB point above the passband.
Q	Loaded quality factor of a tuned circuit. Higher Q = narrower bandwidth, sharper selectivity, but more component sensitivity. PA plate ta